



Manufacturing & Service Operations Management

Publication details, including instructions for authors and subscription information:
<http://pubsonline.informs.org>

Cheap Thrills! An Empirical Analysis of the Impact of Supercenters on Consumer Waste

Suvrat Dhanorkar, Sungho Park, Lina Wang

To cite this article:

Suvrat Dhanorkar, Sungho Park, Lina Wang (2025) Cheap Thrills! An Empirical Analysis of the Impact of Supercenters on Consumer Waste. *Manufacturing & Service Operations Management* 27(5):1604–1624. <https://doi.org/10.1287/msom.2023.0207>

Full terms and conditions of use: <https://pubsonline.informs.org/Publications/Librarians-Portal/PubsOnLine-Terms-and-Conditions>

This article may be used only for the purposes of research, teaching, and/or private study. Commercial use or systematic downloading (by robots or other automatic processes) is prohibited without explicit Publisher approval, unless otherwise noted. For more information, contact permissions@informs.org.

The Publisher does not warrant or guarantee the article's accuracy, completeness, merchantability, fitness for a particular purpose, or non-infringement. Descriptions of, or references to, products or publications, or inclusion of an advertisement in this article, neither constitutes nor implies a guarantee, endorsement, or support of claims made of that product, publication, or service.

Copyright © 2025, INFORMS

Please scroll down for article—it is on subsequent pages



With 12,500 members from nearly 90 countries, INFORMS is the largest international association of operations research (O.R.) and analytics professionals and students. INFORMS provides unique networking and learning opportunities for individual professionals, and organizations of all types and sizes, to better understand and use O.R. and analytics tools and methods to transform strategic visions and achieve better outcomes. For more information on INFORMS, its publications, membership, or meetings visit <http://www.informs.org>

Cheap Thrills! An Empirical Analysis of the Impact of Supercenters on Consumer Waste

Suvrat Dhanorkar,^{a,*} Sungho Park,^b Lina Wang^c

^aScheller College of Business, Georgia Institute of Technology, Atlanta, Georgia 30308; ^bSNU Business School, Seoul National University, Seoul 08826, South Korea; ^cSmeal College of Business, The Pennsylvania State University, University Park, Pennsylvania 16802

*Corresponding author

Contact: sdhanorkar3@gatech.edu,  <https://orcid.org/0000-0003-1350-0867> (SD); spark104@snu.ac.kr,  <https://orcid.org/0000-0001-7201-8769> (SP); lkw5428@psu.edu,  <https://orcid.org/0009-0001-2269-810X> (LW)

Received: April 21, 2023

Revised: June 12, 2024; March 19, 2025

Accepted: May 17, 2025

Published Online in Articles in Advance:
July 14, 2025

<https://doi.org/10.1287/msom.2023.0207>

Copyright: © 2025 INFORMS

Abstract. *Problem definition:* To generate more sales, retailers are incentivized to open large stores. However, large store formats can carry negative environmental externalities. Entry of large stores in a community may stimulate mass consumption through its impact on consumer behaviors and local competition. In this study, we examine the local impact on the amount of consumer waste generated following an expansion of retail Supercenters. *Methodology/results:* We leverage the staggered expansion of Walmart Supercenters and adopt a difference-in-differences approach to investigate the impact of Supercenter entry on consumer waste. Our difference-in-differences estimates suggest that Supercenter entry results in up to a 6.97% increase in consumer waste in the affected counties. The increase is larger for new Supercenter launches compared with Supercenter conversions. In addition, we examine the roles of convenience stores and circular economy channels in mitigating the impact of Supercenter expansion. *Managerial implications:* For policymakers, our results also highlight a silver lining: the negative environmental effect of Supercenter entry can be mitigated through a balanced retail strategy that includes both convenience stores and Supercenters. Furthermore, we show that developing and promoting local circular economy channels may also mitigate the Supercenter effect.

Supplemental Material: The online appendix is available at <https://doi.org/10.1287/msom.2023.0207>.

Keywords: Supercenter expansion • consumer waste • Walmart • convenience stores • circular economy • difference-in-difference

1. Introduction

Today, one of the pressing environmental challenges is the generation of colossal amounts of consumer waste, which is an inadvertent byproduct of embracing mass consumption of goods and services fueled by the growth of retail activities (Vergara and Tchobanoglous 2012, Caro et al. 2020). Among industrialized nations, the United States is one of the largest producers of municipal solid waste (e.g., food waste, packaged goods, disposable goods, used electronics) per capita. In 2018, American consumers, on average, generated 4.9 pounds of waste per person every day (EPA 2020). Although 32% percent of consumer waste is recycled or composted, source reduction is a preferred alternative over recycling and composting, according to the waste management hierarchy proposed by the U.S. Environmental Protection Agency (EPA). Reducing consumer waste improves air and water quality and decreases greenhouse gas emissions that contribute to climate change (Netz et al. 2007). Moreover, it can alleviate the increasing pressure faced by local communities on managing waste using socially and

environmentally sustainable approaches (Tchobanoglous and Kreith 2002).

Research across disciplines in Operations Management (OM), Economics, and Marketing has increasingly examined consumer waste as a behavioral problem, seeking to address this problem by changing consumers' incentives to reduce the generation of consumer waste (Tsiros and Heilman 2005, Homonoff 2018, Akkas and Gaur 2022). One stream of research evaluates incentives that change consumers' purchasing and consumption habits by educating consumers about the environmental impact or by leveraging behavioral nudges (Miranda et al. 1994, Taylor and Todd 1995, Plambeck and Wang 2009, Dhanorkar and Muthulingam 2020, Long and Nasiry 2022). Another burgeoning stream of research examines innovative business models as a channel for postconsumer recovery (Richards and Hamilton 2018, Agrawal et al. 2019, Dhanorkar 2019, Caro et al. 2020, Lowrey et al. 2023). These business models include secondary markets that create substitutes for purchasing primary products and platform-based solutions that allow for efficient matching between demand

and supply that would have traditionally contributed to waste. Although these research streams have established an important link between the generation of consumer waste, consumption, and purchasing behaviors, they have yet to encompass the retail activities as a driver that shapes consumers' behaviors, thereby affecting the amount of waste they produce.

It is well established that retail activities can lead to consequential environmental and societal externalities (Volpe and Lavoie 2008, Caro et al. 2020). For example, retailers are known to increasingly operate larger stores, which often encourage consumers to drive farther to shop than they would have if smaller stores were available nearby (Glaeser 2012). This trend not only substantially increases greenhouse gas emissions due to increased travel (Cachon 2014), but also the large store format induces consumers to purchase larger basket sizes, which contribute to higher waste (Belavina 2021). Walmart is well known as a pioneer in leading large retail store formats. In the retail industry, Walmart's Supercenter expansion has been recognized as one of the most successful and profitable business strategies that motivates its competitors to follow suit. Because of its scale and impact, Walmart's Supercenter expansion has drawn a great deal of attention in scholarly research, which documented a structural change that its expansion has brought to many aspects of local communities, from consumer purchasing behaviors, to health outcomes, to shifts in industry composition and competition (Basker and Noel 2009, Matsa 2011, Bonanno and Goetz 2012). However, the impact of the Supercenter expansion through the lens of sustainability has received limited attention, as indicated in a recent call for research (see, e.g., Volpe and Boland 2022).

In this study, we investigate the impact of Supercenter expansion on consumer waste, broadening our knowledge of the environmental externalities of retail activities that are becoming increasingly important for researchers, policymakers, and managers. To be more specific, we mainly focus on the expansion of Walmart Supercenters, which have successfully expanded throughout the United States, widely influencing shopping and consumption patterns of U.S. consumers. The expansion involves launches of new Supercenters, which average approximately 187,000 square feet in size and offer a wide assortment of general merchandise and grocery goods, featuring a one-stop shopping experience. It also involves conversions of Walmart discount stores, which offer general merchandise and a limited variety of food products and are, on average, about half the size of a typical Supercenter, into Supercenters (referred to as "Supercenter conversions").

To examine the effect of Supercenter expansion on consumer waste, we exploit a quasiexperimental setting that utilizes the staggered expansion of Walmart

Supercenters in the United States (see, e.g., Hwang and Park 2016 and Courtemanche et al. 2019). Using county-level data on consumer waste, we conduct a difference-in-differences analysis of consumer waste in various geographic markets (in four states: California, Florida, Minnesota, and North Carolina) before and after a new Supercenter entry. In particular, we first look for areas where Supercenter openings occurred during our sample period and find comparable areas without Supercenter openings and then estimate the impact of Supercenter openings on the amount of solid waste by comparing the waste amounts between the two areas over time. Next, we investigate whether the impact of Supercenter entry varies with key observed attributes that can potentially provide meaningful policy implications. Specifically, small-format convenience stores have been known to reduce the tendency for overpurchasing due to their limited product assortment (Chenarides et al. 2024). Therefore, we explore the role of small-format convenience stores in ameliorating the impact of Supercenter expansion on waste. Furthermore, the circular economy (CE) has been proposed as a useful approach for mitigating waste and extracting value out of consumer goods (Stahel 2016, Korhonen et al. 2018). For instance, thrift and used merchandise stores could help cycle useful consumer goods between users, and food hubs can help redistribute perishable goods. Similarly, online reuse markets (e.g., Craigslist, Freecycle) have also been shown to promote exchanges of consumer goods (Dhanorkar 2019). We investigate whether the impact of Supercenters on waste amounts is mitigated by the presence of CE channels. Finally, we examine whether the local effects of Supercenter entry vary with income and political inclination—two variables that have been shown to broadly impact environmental attitudes and consumer behaviors.

Our estimates suggest that Supercenter entry results in up to a 6.97% increase in consumer waste in the affected counties, after accounting for various demographic characteristics, time, and county fixed effects. The magnitude of the increase is larger for new Supercenter launches compared with Supercenter conversions. These results are robust to various alternative modeling approaches and specifications. Furthermore, our results show that a greater presence of convenience stores, which are in smaller format sizes than Supercenters, helps ameliorate (by partially offsetting overpurchasing) the impact of Supercenter expansion on consumer waste. Finally, from a policy standpoint, we find that the Supercenter effect can be mitigated (by partially offsetting disposal) through CE channels that promote the exchange and reuse of merchandise and the redistribution of perishable goods. Specifically, we show that online CE channels (e.g., platforms like Craigslist) and offline CE channels for merchandise and perishables (e.g., used merchandise

stores and food hubs) can alleviate the adverse impact of Supercenter expansion on consumer waste.

We organize the remainder of the paper as follows. Section 2 presents a literature review, and Section 3 outlines the development of our hypotheses. We then introduce our empirical approach in Section 4, followed by Section 5, which details the empirical analyses and results. Finally, we discuss our findings and implications and conclude in Section 6.

2. Literature

Our work is related to and builds on the literature on the societal impacts of Supercenter entry, the environmental externalities of the retail supply chain, and the circular economy. Walmart's expansion provides an ideal setting to study the impact of Supercenter entry on consumer behaviors in local communities due to its enormous scale of operation. Walmart is the world's largest company, generating U.S.\$549 billion in revenues and employing 2.2 million workers, according to the Fortune Global 500 in 2020. As such, its expansion is referred to as the most significant shock that leads to a broad structural change in many aspects of local communities (Matsa 2011). These impacts have been investigated in extensive literature, as detailed in Table A1 in the Online Appendix. Most of these studies have primarily focused on economic aspects encompassing Walmart's direct and indirect pricing influence, its impact on competitors, labor market dynamics, and shifts in consumer behavior. However, it is important to note that the examination of Walmart's Supercenter expansion from a sustainability perspective has received limited attention, as highlighted in a recent research call (for reference, see Volpe and Boland 2022). Within this sustainability framework, prior research has examined the effects of Supercenter entry on consumer health outcomes. Nevertheless, our study contributes novel insights to the field, being the first to specifically assess the influence of Supercenter entry on consumer waste generation.

The second literature stream studies the environmental externalities of the retail supply chain. Belavina et al. (2017) compare environmental costs between per-order and subscription revenue models offered by an online retailer where the former induces consumers to purchase larger basket sizes and the latter incentives more frequent purchases. They find the environmental costs of food waste due to larger basket sizes significantly exceed the environmental costs of delivery-related emissions due to frequent purchases. Waste in the retail supply chain has increasingly spurred scholarly attention because it contributes to a significant portion of municipal solid waste (see Akkaş and Gaur 2022 for a review). One important driver of waste in the retail industry comes from the large amounts of

edible produce that fail to pass the aesthetic bar and are therefore thrown away (Richards and Hamilton 2018). Another important driver is the aging of inventory, which is attributable to a retailer's inventory management decisions (e.g., carrying large safety stock to meet high service levels) (Akkaş and Honhon 2022). Looking beyond store waste, Belavina (2021) evaluates the impact of store density on market waste comprising both store waste and consumer waste. The author argues that improved access to stores reduces market waste because a reduction in consumer waste as a result of consumers purchasing a smaller basket size well compensates for an increase in store waste due to more decentralized inventory. Additionally, Cachon (2014) identifies that larger retail store formats increase carbon emissions by prompting consumers to travel greater distances. Our study extends this body of literature by examining the environmental externalities of the retail supply chain following the expansion of Supercenters and exploring the role of smaller convenience stores in mitigating these effects.

The third area of related work includes studies looking at the role of the circular economy in facilitating product reuse through transaction or exchange. Product reuse is a preferred alternative over other initiatives, such as recycling, energy recovery, and disposal, according to the EPA's Waste Management Hierarchy. The CE activities create substitutes for primary products, thereby reducing the purchase of new products (Suslow 1986). Evidence of CE cannibalization of new product demand is documented in literature from automobiles to consumer electronics to fashion products (Waldman 2003, Ghose et al. 2005, Chen et al. 2013, Alev et al. 2020). In addition to CE channels for nonperishable goods, circular economy for perishable goods (e.g., food banks) is found to reduce waste by diverting from landfills wasted perishable products, which traditionally contribute to a large portion of municipal solid waste (Akkaş and Gaur 2022). Our study provides empirical evidence on the role of the different types of secondary markets in mitigating the waste problem arising from Supercenter entry.

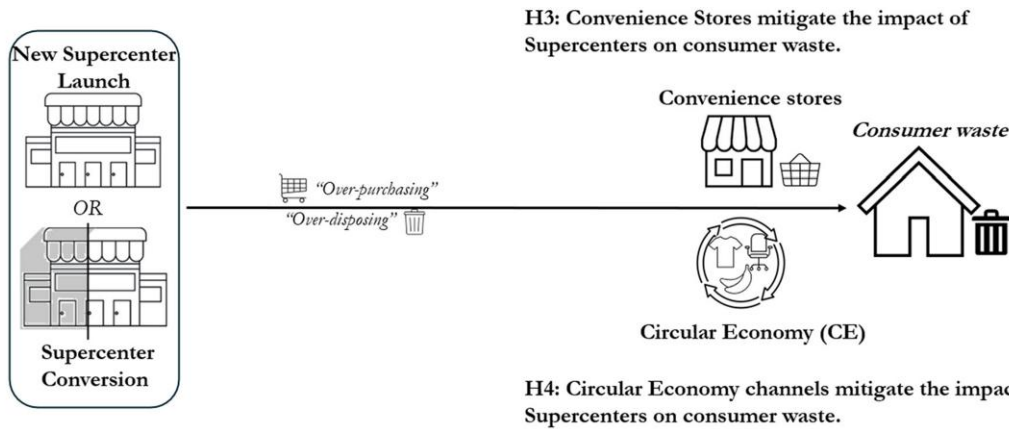
3. Hypotheses

In this section, we investigate the underlying mechanisms through which a Supercenter entry influences consumer waste. Additionally, we examine the mitigating role of convenience stores and circular economy channels on this Supercenter effect. We develop a set of hypotheses, as illustrated in Figure 1.

We argue that Supercenter entry leads to higher consumer waste due to a rise in overpurchasing behaviors. First, Supercenters' arrival in communities significantly alters consumer behaviors, predominantly

Figure 1. Research Framework

H1: Supercenter entry leads to an increase in consumer waste.
H2: New Supercenter launch leads to higher consumer waste compared to Supercenter conversions.



Note. H, hypothesis.

due to Walmart’s “Everyday Low Price” (EDLP) strategy, which is integral to its business model, especially in its Supercenters. This pricing strategy ensures that prices offered at Supercenters are substantially lower than prices in incumbents’ stores (Singh et al. 2006, Ailawadi et al. 2010). In addition to underselling incumbents, Supercenters’ entry pressures incumbents to react by reducing their own prices (Basker and Noel 2009). Thus, Supercenters’ entry lowers the overall price levels faced by consumers, which results in a new market equilibrium that affords consumers a larger basket size given the same income (Hausman and Leibtag 2007). There is ample empirical evidence suggesting that retail price reduction following Supercenters’ entry induces overpurchasing by consumers (Basker 2005, Volpe and Lavoie 2008).

Second, further contributing to the overpurchasing phenomenon is the Supercenter’s large store format and its typically remote location, away from residential areas. Walmart Supercenters, pioneering the big-box store concept in the United States, offer a comprehensive one-stop shopping experience but are often situated farther from residential areas than smaller store formats (Cachon 2014). Given the greater distance and effort required by consumers to reach a Supercenter, they maximize the utility of each visit by purchasing larger basket sizes. Moreover, to avoid the inconvenience of making another lengthy journey soon after, consumers may purchase more than what’s needed. Consequently, this behavior contributes to an increase in consumer waste, as the surplus products purchased on these infrequent, bulk-buying trips are more likely to go unused or expire (Belavina 2021). Therefore, we hypothesize that the presence of Supercenters leads to an increase in consumer waste.

Hypothesis 1. *Supercenter entry leads to an increase in consumer waste.*

The impact of Supercenters on consumer waste may vary, depending on the type of entry. Walmart Supercenters enter local communities either by converting existing discount stores into Supercenters (i.e., conversion) or by establishing entirely new ones (i.e., launch). The conversion involves expanding the size of an existing store to add a comprehensive grocery section featuring fresh produce, meats, bakery items, deli products, and seafood. This expansion aims to increase consumer demand in food categories, potentially increasing sales in other categories. However, the impact on consumer purchasing behavior is expected to be less pronounced for conversions than for new Supercenter launches because the former primarily expands product offerings of food items. Hwang and Park (2016) found that Supercenter conversions led to a 41% rise in average weekly household expenditure, yet the demand spillover from the expanded food categories to nonfood categories was limited. Consequently, the behavioral influence of converting existing stores on consumers is likely less significant than that of establishing new Supercenters, which expand across a full range of merchandise. Moreover, because Supercenter conversions focus primarily on food, they might only influence price reductions in food categories, having a minimal effect on other categories out of the scope of its expansion (Ailawadi et al. 2010). In contrast, new Supercenters, with their comprehensive merchandise expansion, could exert a more pronounced influence on consumer overpurchasing behaviors and, consequently, also lead to higher waste generation. Thus, we hypothesize that the introduction of new Supercenters contributes to

greater consumer waste compared with the conversion of existing stores into Supercenters.

Hypothesis 2. *New Supercenter launch leads to higher consumer waste compared with Supercenter conversions.*

In contrast to Supercenters, traditional convenience stores (e.g., 7-Eleven, QuikTrip, Wawa, and other non-chain neighborhood stores commonly seen in the United States) are likely to generate very different shopping behaviors. First, convenience stores tend to encourage consumers to purchase smaller basket sizes compared with Supercenters. Their proximity to residential areas reduces the cost and effort associated with shopping trips, enabling consumers to shop more frequently but in smaller quantities (Martin et al. 2020). This shopping behavior is in stark contrast to the bulk-buying practices encouraged at Supercenters, which typically result in elevated household inventory levels (Aschemann-Witzel et al. 2016, 2017). Additionally, the smaller packaging sizes offered at convenience stores, as opposed to the larger options at Supercenters, further promote smaller basket sizes and help maintain lower inventory levels at homes (Williams et al. 2020).

Second, convenience stores also play a role in reducing the tendency for overpurchasing due to their limited product assortment (Chenarides et al. 2024). This narrower selection reduces the cognitive load on consumers, enabling more thoughtful and intentional purchasing decisions (Iyengar and Lepper 2000, Chernov 2003). The absence of an overwhelming array of choices helps prevent the impulsive and excessive purchasing often encountered in Supercenters (Hausman 2000). The store design and layout, which do not typically promote impulse buying as in Supercenters, further support mindful shopping (Farley and Ring 1966, Flamand et al. 2023). This environment further encourages more thoughtful purchasing, aligning with the trend toward more mindful consumption and potentially reducing unnecessary purchases. These elements combine to cultivate shopping patterns marked by smaller basket sizes and deliberate purchasing, contributing significantly to waste reduction (Belavina 2021).

Finally, it is known that the unit price of a product sold at convenience stores is generally higher than when sold at other major retail channels, such as grocery stores or big-box retailers (Leibtag 2006). As a result, consumers may feel obligated to fully use (rather than waste) the purchased products to avoid the guilt associated with wasting money spent on the purchase. This phenomenon, known as the sunk-cost effect, intensifies with higher expenditures (Gourville and Soman 2002). To sum up, we posit that a higher presence of convenience stores in a region will partially offset the Supercenter effect.

Hypothesis 3. *Convenience stores mitigate (through better purchase behaviors) the negative impact of Supercenters on consumer waste.*

In the context of growing environmental concerns, the circular economy offers a promising framework to reduce waste and extract value out of existing consumer goods (Stahel 2016, Korhonen et al. 2018). The CE concept advocates for the reuse of goods that retain value and the recycling of resources from products at the end of their useful life (Schulze 2016). Thus, CE activities—offline and online—promote the circulation of consumer goods, thereby promoting sustainable consumption and waste reduction. For instance, thrift and used merchandise stores facilitate the recirculation of durable consumer goods, while food hubs help redistribute perishable goods to reduce waste. Similarly, online reuse markets (e.g., Craigslist, Freecycle, Facebook Marketplace) have also been shown to promote exchanges of consumer goods (Dhanorkar 2019). Therefore, the greater presence of CE alternatives should keep the goods in circulation longer and thus mitigate the Supercenter effect on consumer waste.

Furthermore, CE channels may also have a broader influence on consumer behaviors. Although CE channels undoubtedly offer a more sustainable alternative for disposing of old and acquiring new products, such markets also offer a superior hedonic experience, where consumers enjoy searching, bargaining, the festive atmosphere, and social interactions (Belk et al. 1988, Sherry 1990). The presence of CE channels in communities not only improves the efficiency of peer-to-peer exchanges, but also significantly influences consumer behaviors. Moving away from the “buy new” mentality often linked with Supercenter shopping, CE channels promote and foster a culture of reuse. This reuse mindset is shown to reduce demand for new products across multiple industries, including automotive, electronics, and fashion, as supported by various studies (Waldman 2003, Ghose et al. 2005, Chen et al. 2013, Alev et al. 2020). Empirical evidence suggests that CE channels can significantly reduce solid waste in local communities (Fremstad 2017, Dhanorkar 2019). By encouraging the sale or donation of items instead of disposal, CE channels help mitigate the environmental impact of Supercenters on consumer waste. Thus, we hypothesize that CE channels, through encouraging and facilitating better disposal behaviors, mitigate the negative impact of Supercenters on consumer waste.

Hypothesis 4. *CE channels mitigate (through better reuse/disposal behaviors) the negative impact of Supercenters on consumer waste.*

4. Empirical Approach

4.1. Data and Variables

We use a county-level panel data set to examine the longitudinal relationship between Supercenters and consumer waste. To construct our data set, we rely on a variety of information sources. First, we collected consumer waste data from state environmental agencies for four states—California, Florida, Minnesota, and North Carolina. Second, we used information on Walmart's Supercenter entry to create the quasi-experimental setup. Third, we obtained a variety of county-level demographic data (e.g., population, wealth, age) from Sage Business Statistics compiled by Woods & Poole Economics Inc. In addition, we obtained data on internet connectivity from the Federal Communications Commission. Finally, we obtained data on housing prices from the Federal Housing Finance Agency. We describe the main variables below.

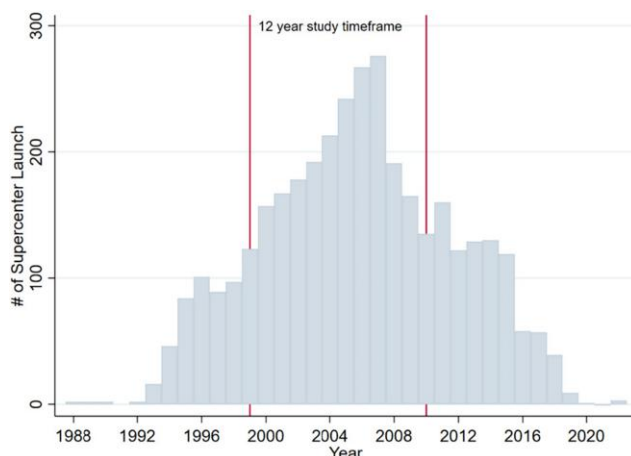
To identify the Walmart effect on consumer waste, we focus on the 1999–2010 time period. We do this for five reasons. First, as indicated in earlier research (Dhanorkar 2019), few states report detailed county-level consumer waste data. Even for those few states that do report such data (e.g., California, Florida, Minnesota, North Carolina), good-quality county-level waste data were not readily available before 1999. This data limitation drives the starting point of our sample period. Second, Walmart's Supercenter expansion happened in epochs (see Figure 2), and the 1999–2010 timeframe roughly represents one of the three epochs. During this 12-year period, the number of Walmart Supercenters increased by 380%, with an average 16% year-over-year growth rate. In comparison, for example, during the following 12-year period (i.e., between 2011 and 2022), the number of Supercenters increased by only 30%, with an average 1.9% year-over-year growth rate.¹ During the 1999–2010 timeframe, several counties in the four states considered in this study

also experienced their first-ever Walmart Supercenter entry.² In addition, for each county that received a Supercenter, we are also able to identify multiple comparable counties within the same state that had not yet seen a Supercenter entry. This allows us to cleanly identify the Supercenter entry effect at the county level using a matched sample approach. Fourth, during our sample period, the percentage of e-commerce sales remained less than 5% of the total retail sales.³ Thus, the expansion of online retail, which primarily took place after 2010, is unlikely to confound our results for Walmart's effect. This logic is in line with several recent studies that have examined various outcomes related to Walmart's expansion (Matsa 2011, Hwang and Park 2016, Arcidiacono et al. 2020). Finally, one of the objectives of this study is to examine the potential of CE channels in mitigating the Supercenter effect. Primary expansion of one of the first online CE channels (i.e., Craigslist) in the United States also happened during the mid-2000s. To implement our identification strategy, we therefore specifically focus on the 1999–2010 timeframe.

4.1.1. Consumer Waste. The Environmental Protection Agency defines "Municipal Solid Waste (MSW)—more commonly known as trash or garbage—to consist of everyday items we use and then throw away, such as product packaging, grass clippings, furniture, clothing, bottles, food scraps, newspapers, appliances, paint, and batteries" (EPA 2016). Over the years, the EPA has conducted regular MSW characterization studies (see, e.g., EPA 2011, 2013, 2014), which have helped to authenticate the concept of MSW and created consistency in the way MSW information is recorded.

We conduct our econometric analysis at the county level. To our knowledge, there is no central repository of county-level consumer-generated MSW data. To identify these data sources, one of the authors conducted an extensive survey of state environmental agencies, followed by specific conversations with state officials. Finally, we were able to obtain detailed county-level consumer waste data from four U.S. states over the sample. For California, we collected county-level MSW data for 58 counties from the Data Reporting System managed by CalRecycle.⁴ For Florida, we collected data for the 67 counties from the Florida Department of Environmental Protection.⁵ For Minnesota, we collected data on 87 counties from the Minnesota Pollution Control Agency.⁶ Finally, for North Carolina, we extracted the data from annual reports for its 100 counties from the North Carolina Department of Environmental Quality.⁷ We log transform the dependent variables to reduce skew and to allow inference as a percentage change (Greenwood and Agarwal 2016). Subsequently, we show robustness by using alternative measures of consumer waste, normalized by various demographic characteristics, such as population,

Figure 2. (Color online) Walmart Supercenter Expansion



households, and area. This approach accounts for a county's inherent potential to generate waste.

4.1.2. Control Variables. To account for various time-varying factors that may influence consumer waste, we include multiple control variables. We control for housing price index (HPI), as it reflects real estate prices that affect the size of homes and thus resource consumption within a county. By controlling HPI, we aim to partial out this effect, allowing us to more precisely identify the impact of the Supercenter entry on consumer waste. HPI data were gathered for metropolitan areas, which were then mapped to corresponding counties to create a measure of HPI for each county. We obtained these data from the Federal Housing Finance Agency. We control for the median age of residents in the county because environmental attitudes toward consumption and recycling may be correlated with an individual's age. We control for the average level of college education in the county because the residents' consumption and disposal patterns may be at least partly connected to an individual's level of education. Education data were obtained from the U.S. Census and are calculated as the percent of county residents with a college degree. We control for wealth because it could also strongly correspond with consumption patterns. The Woods & Poole Wealth Index is used to capture wealth within a county (Dhanorkar and Muthulingam 2020). The index is a better measure of wealth compared with other metrics, such as average income, because it captures relative personal income per capita, weighted by the source of income. Thus, the average wealth index in the United States is 100; values above (below) 100 indicate regions that are wealthier (poorer) than the national average. These data were obtained from Sage Stats. Descriptive statistics are provided in Table 1,

and correlations are included in Table A3 of Online Appendix A. As indicated in Table 1, several of the control variables were log-transformed to reduce skew. In addition to these time-varying controls, we use county fixed effects to control for factors within counties that could influence MSW trends. We also use time fixed effects for each calendar year to account for temporal shocks.

4.1.3. Moderating Variables. To assess the moderating effect of convenience stores, we collected data from the U.S. Census Bureau on convenience stores (North American Industry Classification System (NAICS) 445120) in each county. According to the U.S. Census Bureau, "...this industry comprises establishments known as convenience stores or food marts (except those with fuel pumps) primarily engaged in retailing a limited line of goods."⁸ We use the number of stores under the industry classification code in each county to measure the presence of *convenience stores*.

To evaluate the moderating effects of circular economy channels, we augmented our data set with new variables for a variety of CE channels. First, traditional offline CE channels, such as thrift and used merchandise stores, could help cycle useful consumer goods between users. In addition to the secondary markets that circulate general merchandise, CE channels like food banks may also play a crucial role by diverting from landfills wasted perishable products which traditionally contribute significantly to municipal solid waste (Akkaş and Gaur 2022). More recently, online reuse markets have emerged as powerful platforms that facilitate the exchange of goods on a broader scale (Fremstad 2017, Dhanorkar 2019). Reuse platforms (e.g., Craigslist) connect individuals for the sale or donation of used or surplus items, fostering a culture of reuse and reducing waste accumulation. First, we

Table 1. Descriptive Statistics

Variables	Treated group		Control group	
	Mean	Std. Dev	Mean	Std. Dev
MSW (log)	11.9100	1.4186	10.1613	1.7402
MSW per capita (log)	0.7738	0.1905	0.6504	0.2652
MSW per household (log)	1.3770	0.2603	1.1892	0.3785
MSW per area (log)	5.3654	1.3099	3.6432	1.5190
Median Age	38.5018	5.1277	40.2045	4.8481
Education	33.4981	6.8222	31.8109	5.4114
Online Connectivity (log)	2.0987	0.6622	1.8257	0.5605
Housing Price Index (log)	5.5843	0.5580	5.4629	0.6171
Wealth Index (log)	4.4502	0.2121	4.4183	0.2366
Supercenter	0.6979	0.4593	0.0000	0.0000
Supercenters (#)	0.8905	0.6950	0.0000	0.0000
Convenience Stores (log)	3.6357	1.2671	2.5578	1.1783
CE channel (Merchandise) (log)	2.1109	1.3134	1.1294	1.1809
CE channel (Perishable) (log)	0.6977	0.8598	0.4387	0.7206
CE channel (Online)	0.1444	0.3516	0.0722	0.2589

collect the number of used merchandise stores—that is, surplus and thrift stores (NAICS 453310)—from the U.S. Census Bureau. This serves as a proxy for the *CE channel (Merchandise)*. We also collected the number of food banks—that is, food banks, meal delivery programs, and soup kitchens (NAICS 624210); we use this as a proxy for the *CE channel (Perishable)*. To access the *CE channel (Online)*, we followed earlier research (Fremstad 2017, Dhanorkar 2019) and collected data on Craigslist entry into specific counties in our data set.

4.2. Difference-in-Differences Strategy

To estimate the effect of Supercenters on consumer waste in geographic markets, we exploit a quasi-experimental setting: the new launch of or conversion to Walmart Supercenters (from Walmart Discount Stores) in the United States. To this end, we follow recent research that leverages Walmart Supercenter entry in the United States to examine various outcomes such as shopping behaviors (Hwang and Park 2016) and obesity (Courtemanche and Carden 2011).

Our main intention is to analyze differences in consumer waste as a function of Supercenter entry, as well as other time-variant and time-invariant factors. Given the nature of our consumer waste data, we conduct our main analysis at the county level, following Dhanorkar (2019). Thus, we compare the differences in consumer waste for geographic markets (i.e., counties in various states) before and after a new Supercenter entry. For our analysis of the treatment effect (i.e., Hypothesis 1), we employ a regression-based difference-in-differences approach, as specified in Equations (1) and (2), to examine the impact of the first-ever Supercenter entry and the ongoing presence of Supercenters, respectively:

$$\ln(W_{st}) = \alpha \cdot C_s + \gamma \cdot H_t + \pi \cdot X_{st} + \beta \cdot \text{Supercenter}_{st} + \varepsilon_{st}, \quad (1)$$

$$\ln(W_{st}) = \alpha \cdot C_s + \gamma \cdot H_t + \pi \cdot X_{st} + \beta \cdot \text{Supercenters}(\#)_{st} + \varepsilon_{st}, \quad (2)$$

where W_{st} is the dependent variable for region s at time t ; C_s is a vector of county fixed effects; and H_t is a vector of year fixed effects. Thus, the county fixed effects control for time-invariant differences across regions, and the year fixed effects control for common macroeconomic shocks across time. Additional time-varying control variables X_{st} are included in the models to account for factors that vary within each region over time. To ensure that our findings are robust to alternative operationalizations of the dependent variable, we present results using four measures. To this end, we reproduce the results using a nonstandardized dependent variable—*Consumer Waste*—as well

as dependent variables standardized by the population, households, and county area, respectively.

To code the treatment variable, we collected data on all Walmart stores. This data set contains information on the address, store number (store ID), store type (i.e., Supercenter or discount store), opening date (or conversion date if the store is a Supercenter converted from a discount store), and the U.S. county of location for each Walmart store. Conversions account for up to two-thirds of the total Supercenter entries. The data up to 2006 were already captured and made available by Holmes (2011). Subsequently, we extended these data by collecting new information up to 2010 based on the announcements on Walmart's website. Thus, for our analysis, we consider the 12-year timeframe between 1999 and 2010 when Walmart's primary expansion of Supercenters took place. In Equation (1), Supercenter_{st} is the binary indicator for the first-ever entry of a Walmart Supercenter. It takes the value of one in all subsequent years after the launch of a new Supercenter or conversion, and zero otherwise. This variable is utilized to examine the impact of Supercenters as proposed in Hypothesis 1. In Equation (2), $\text{Supercenters}(\#)_{st}$ accounts for the presence of multiple Walmart Supercenters [$\text{Supercenters}(\#)_{st} = n$ in all the years following new Supercenter entries or conversions, and zero otherwise]. The coefficients β are the difference-in-differences estimates of the effect of the ongoing presence of Supercenters on the dependent variable. Following the approach in Dhanorkar (2019), we consider the entire county as being "treated" when a Supercenter is launched in that county.

To compare the effects of a Supercenter's launch versus conversion (i.e., Hypothesis 2), we use Equation (3). $\text{Supercenter_New}_{st}$ denotes the binary indicator for a new Walmart Supercenter launch (with a value of one in all postlaunch years and zero otherwise), and $\text{Supercenter_Conv}_{st}$ denotes the binary indicator for a Walmart Supercenter conversion (with a value of one in all postconversion years and zero otherwise) from a traditional Walmart discount store to a Walmart Supercenter.

$$\ln(W_{st}) = \alpha \cdot C_s + \gamma \cdot H_t + \pi \cdot X_{st} + \beta_1 \cdot \text{Supercenter_New}_{st} + \beta_2 \cdot \text{Supercenter_Conv}_{st} + \varepsilon_{st}. \quad (3)$$

To test the moderation hypotheses (i.e., Hypothesis 3 and Hypothesis 4), we use Equations (4) and (5), where we include interaction terms between the treatment (i.e., either Supercenters_{st} or $\text{Supercenters}(\#)_{st}$) and the moderator M_{st} . For Hypothesis 3, studying the convenience store moderator (i.e., Hypothesis 3), M_{st} represents the number of convenience stores. To reduce skew, we use the log-transformed number of convenience stores in the county. For Hypothesis 4, that examines three aspects of CE channels, when studying the CE

channel (*Merchandise*) and *CE channel (Perishable)*, M_{st} is the log-transformed number of thrift stores and log-transformed number of food banks, respectively. When studying *CE channel (Online)*, M_{st} is a binary variable for Craigslist's entry. In alignment with prior studies (e.g., Chan et al. 2016, Greenwood and Agarwal 2016, Dhanorkar 2019), we code the *CE channel (Online)* as one in all of the years following Craigslist entry into the county, and zero otherwise. Following research on the difference-in-differences technique (Bertrand et al. 2004), we presented results using unit (i.e., county) and time (i.e., year) fixed effects; we log transform the dependent variable to reduce skew and to allow inference as a percentage change (Greenwood and Agarwal 2016).

$$\ln(W_{st}) = \alpha \cdot C_s + \gamma \cdot H_t + \pi \cdot X_{st} + \beta \cdot \text{Supercenter}_{st} \cdot M_{st} + \text{Supercenter}_{st} + M_{st} + \varepsilon_{st}, \quad (4)$$

$$\ln(W_{st}) = \alpha \cdot C_s + \gamma \cdot H_t + \pi \cdot X_{st} + \beta \cdot \text{Supercenters}()_{st} \cdot M_{st} + \text{Supercenters}()_{st} + M_{st} + \varepsilon_{st}. \quad (5)$$

4.3. Propensity Score Matching

The difference-in-difference technique relies on the exogeneity of treatment, which may be violated because Walmart is likely to selectively launch Supercenters in certain locations based on demographic factors. Walmart's decisions to propose supercenter locations are largely influenced by factors such as population density, wealth, and employment levels. Previous research on Walmart's market entry highlights the importance of population size (e.g., Jia 2008, Zhu and Singh 2009). Moreover, recent studies suggest that Walmart often targets more rural areas, leading us to include population density as a covariate to account for both population and rurality effects. Additionally, Walmart is known to prioritize lower-income markets (Halebsky 2009). Consequently, wealth and employment levels are included as covariates to capture the impact of income on Walmart's entry decisions.

Despite Walmart's decision to enter a county, the actual timing of Supercenter launch appears to involve significant randomness. Walmart often selects several potential locations, some of which may face delays or obstacles, such as zoning issues, that are beyond Walmart's control. For example, in Eureka, CA, Walmart proposed opening a Supercenter in 1999, but the project was blocked due to zoning regulations. As a result, Walmart did not open its first Supercenter in Eureka until 2012. Similarly, in Alachua, FL, Walmart proposed opening a Supercenter in 2004, but the chosen location was initially rejected. It took four years for Walmart to establish its first Supercenter there in 2008.

These examples illustrate the substantial random variation in the timing and location of Walmart's

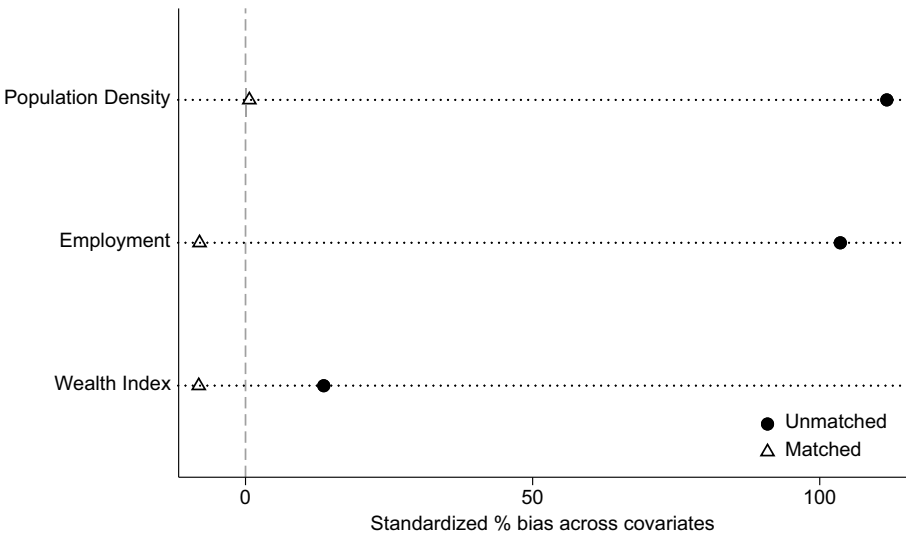
Supercenter openings. In our study, counties like Eureka and Alachua—where Walmart's proposals faced delays or obstacles—serve as comparable control groups to counties where proposals were approved without significant delays. Please recall that our control group consists of counties where Supercenters have not yet opened but are expected to open eventually, providing an ideal comparison for analyzing the treatment effects.

To address potential selection bias due to any potential differences across the treated and untreated counties and to address changes in these differences over time, we apply a dynamic propensity score matching (PSM) technique (Rosenbaum and Rubin 1983). When applied correctly, this approach should provide matched control counties (i.e., those without Walmart Supercenters) that are similar to the treated counties (i.e., those that have experienced Walmart Supercenter entry), with respect to the observed characteristics on a year-over-year basis (Heckman et al. 1997).

To implement PSM, we use the nearest three neighbors (NN3) algorithm. In the Online Appendix, we use alternative matching methods, including one-to-one matching without replacement, Coarsened Exact Matching, and Mahalanobis matching, based on the same set of matching covariates as in NN3 matching. We matched the treated and untreated counties based on three county-level covariates—population density, employment, and wealth—using a caliper size of 0.2. These covariates should theoretically influence Supercenter entry into the county (Ellison et al. 2010); as expected, we find all three covariates to significantly predict Supercenter entry. Specifically, the NN3 matching process significantly improved the balance in means (across treated and control units) for matched covariates.

The PSM approach resulted in a significant reduction in both mean bias and median biases, with the mean bias dropping from 76.3 to 5.6 and the median bias from 103.6 to 8.0. Prior research (e.g., Rosenbaum and Rubin 1985, Rubin 2001) suggests that, following PSM, the resulting bias “B” (i.e., the absolute standardized difference of the means of the linear index of the propensity score in the treated and control group) be less than 25 and variance ratio “R” (the ratio of treated to control variances of the propensity score index) be between 0.5 and 2, for the samples to be considered sufficiently balanced. Our matching approach resulted in $B=14.4$ and $R=0.51$, thus satisfying the conditions for a well-balanced sample. Notably, although minor differences between the treated and control samples are expected for individual covariates, even after PSM (Brynjolfsson et al. 2011, Xu et al. 2017), our matching approach resulted in the lack of any statistical difference between pretreatment trends for MSW between treated and control groups. We also present balance statistics for the matching covariates in Figure 3 and a comparative boxplot analysis of the

Figure 3. Balance Statistics



treated and untreated groups across these covariates both prematching and postmatching in Figure 4. These figures indicate that the matching process has effectively balanced the covariates between the groups, thereby enhancing the reliability of comparisons between them.

5. Analysis

5.1. Quantifying the Supercenter Effect

To evaluate Hypothesis 1, we use model specifications shown in Equations (1) and (2) to estimate the impact of Supercenter entry on MSW, while accounting for other time-variant and time-invariant factors. Using this approach, we obtained an $R^2 > 80\%$ for all models. Table 2 shows the results for the main models with MSW, MSW per Area, MSW per Capita, and MSW per Household as the dependent variables. Notably, because the dependent variables are log-transformed,

we can roughly interpret the coefficient estimates as percentage changes in the dependent variable. Moreover, two dependent variables, MSW per Capita and MSW per Household, directly account for temporal changes in population in each area. We generally find that Supercenter entry leads to a statistically significant increase in consumer-generated waste (results shown in Table 2, columns (1), (3), (5), and (7)). Specifically, we find that Supercenter entry leads to a roughly 6.97% increase in MSW; a 2.24% increase in MSW per capita; and a 3.23% increase in MSW per Household. The smaller magnitude of increase in MSW when normalized by capita and household suggests that the influence of Supercenter entries is not uniformly experienced across all individuals and households in the county. The Supercenter entry may have a more pronounced effect on those living relatively closer to the

Figure 4. (Color online) Boxplots of Treated and Untreated Groups



Table 2. Main Effect—Impact of Supercenter Entry

Variables	(1) MSW (raw)	(2) MSW (raw)	(3) MSW (per capita)	(4) MSW (per capita)	(5) MSW (per Household)	(6) MSW (per Household)	(7) MSW (per Area)	(8) MSW (per Area)
<i>Median Age</i>	−0.0048 (0.0084)	−0.0030 (0.0083)	0.0081*** (0.0031)	0.0087*** (0.0031)	0.0062 (0.0043)	0.0070 (0.0043)	−0.0022 (0.0076)	−0.0004 (0.0075)
<i>Education</i>	−0.0006 (0.0009)	−0.0006 (0.0009)	−0.0004 (0.0005)	−0.0004 (0.0005)	−0.0007 (0.0006)	−0.0007 (0.0006)	−0.0008 (0.0009)	−0.0008 (0.0009)
<i>Online Connectivity</i>	−0.0486*** (0.0158)	−0.0538*** (0.0157)	−0.0272*** (0.0066)	−0.0289*** (0.0066)	−0.0397*** (0.0093)	−0.0421*** (0.0093)	−0.0543*** (0.0138)	−0.0592*** (0.0138)
<i>Housing Price Index</i>	0.0764** (0.0351)	0.0737** (0.0352)	0.0469*** (0.0141)	0.0460*** (0.0141)	0.0698*** (0.0201)	0.0684*** (0.0201)	0.0978*** (0.0294)	0.0954*** (0.0294)
<i>Wealth Index</i>	0.4150*** (0.1134)	0.4228*** (0.1128)	0.1900*** (0.0479)	0.1924*** (0.0478)	0.2398*** (0.0671)	0.2432*** (0.0670)	0.4035*** (0.1086)	0.4110*** (0.1079)
<i>Supercenter</i>	0.0697*** (0.0169)		0.0224*** (0.0067)		0.0323*** (0.0094)		0.0633*** (0.0150)	
<i>Supercenters (#)</i>		0.0557*** (0.0115)		0.0174*** (0.0050)		0.0248*** (0.0069)		0.0521*** (0.0106)
Constant	9.7845*** (0.6973)	9.7038*** (0.6912)	−0.6277** (0.2670)	−0.6521** (0.2666)	−0.2716 (0.3740)	−0.3062 (0.3733)	3.0056*** (0.6530)	2.9279*** (0.6473)
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,872	2,872	2,872	2,872	2,872	2,872	2,872	2,872
R ²	0.980	0.980	0.831	0.831	0.835	0.835	0.987	0.987

Notes. Robust Standard Errors are in parentheses. Columns (1) and (2) show results for raw MSW. Columns (3)–(8) show results for MSW normalized by population, households, and area respectively. Treated and control counties are matched using PSM on population density, employment, and wealth.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

store; the effect may also vary depending on the demographic factors, as shown in Tables B4.a and B4.b in Online Appendix B. Finally, the results show that *Supercenter* entry leads to an approximately 6.33% increase in MSW per area. Overall, we find substantial support for our main proposition, which argues that Supercenters increase consumer waste. We find similar effects with an alternative specification that uses the number of *Supercenters* (results shown in Table 2, columns (2), (4), (6), and (8)), indicating that each additional Supercenter has an increasingly detrimental effect on consumer waste.

Our analysis of contextual effects hinges on the validity of the main treatment effect (i.e., Hypothesis 1). Before moving on to additional analysis (i.e., Hypotheses 2–4), we conduct a variety of checks (e.g., alternative matching, leads-lags model, falsification checks, controls

without log transformation) to ensure the robustness and validity of the main treatment effect of Supercenter expansion. For brevity, these robustness checks are summarized in Table 3, and the detailed results are provided in Online Appendix B.

5.2. Supercenter Launch vs. Conversion

Given the validity and robustness of the main treatment effect, we now test H Hypothesis 2. Here, we analyze whether and how the two approaches of Supercenter entry—namely, new Supercenter launches versus Supercenter conversions—differently affect consumer waste. We use the model specification shown in Equation (3) to examine their differential impact. Table 4 reports the estimation results across the measurement of consumer waste, including MSW, MSW per Area, MSW per Capita, and MSW per Household.

Table 3. Summary—Evaluating Robustness of the Treatment Effect

Robustness test	Location
Alternative Matching: 1-to-1 PSM	Online Appendix Table B1.a
Alternative Matching: Coarsened Exact Matching	Online Appendix Table B1.b
Alternative Matching using additional covariates: All Walmart Stores	Online Appendix Table B1.c
Alternative Matching using additional covariates: All Superstores	Online Appendix Table B1.d
Excluding Control Counties Neighboring the Treated County	Online Appendix Table B1.e
Alternative Treatment: Walmart & Costco supercenters	Online Appendix Table B2
Dynamic Treatment Effect Using Doubly Robust DID Estimator	Online Appendix Figure B1
Placebo Treatment and Shuffle Tests	Online Appendix Figure B2
Control variables without Log-transformation	Online Appendix Table B3

Table 4. Results—New Supercenter Launches vs. Supercenter Conversions

Variable	MSW			MSW (per capita)			MSW (per Household)			MSW (per Area)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Median Age	−0.0048 (0.0084)	−0.0051 (0.0086)	−0.0053 (0.0085)	0.0081*** (0.0031)	0.0080** (0.0032)	0.0079** (0.0031)	0.0062 (0.0043)	0.0060 (0.0043)	0.0059 (0.0043)	−0.0022 (0.0076)	−0.0024 (0.0078)	−0.0026 (0.0077)
Education	−0.0006 (0.0009)	−0.0005 (0.0009)	−0.0006 (0.0009)	−0.0004 (0.0005)	−0.0004 (0.0005)	−0.0004 (0.0005)	−0.0007 (0.0006)	−0.0007 (0.0006)	−0.0007 (0.0006)	−0.0008 (0.0009)	−0.0007 (0.0009)	−0.0008 (0.0009)
Online Connectivity	−0.0486*** (0.0158)	−0.0457*** (0.0161)	−0.0452*** (0.0159)	−0.0272*** (0.0066)	−0.0261*** (0.0066)	−0.0259*** (0.0065)	−0.0397*** (0.0093)	−0.0380*** (0.0093)	−0.0378*** (0.0092)	−0.0543*** (0.0138)	−0.0521*** (0.0141)	−0.0516*** (0.0138)
Housing Price Index	0.0764** (0.0351)	0.0700** (0.0357)	0.0773** (0.0352)	0.0469*** (0.0141)	0.0449*** (0.0142)	0.0473*** (0.0142)	0.0698*** (0.0201)	0.0669*** (0.0202)	0.0703*** (0.0202)	0.0978*** (0.0294)	0.0918*** (0.0297)	0.0985*** (0.0295)
Wealth Index	0.4150*** (0.1134)	0.3934*** (0.1168)	0.3992*** (0.1141)	0.1900*** (0.0479)	0.1820*** (0.0482)	0.1838*** (0.0476)	0.2398*** (0.0671)	0.2282*** (0.0678)	0.2308*** (0.0669)	0.4035*** (0.1086)	0.3855*** (0.1117)	0.3908*** (0.1091)
Supercenter Conv.	0.0697*** (0.0169)		0.0686*** (0.0166)	0.0224*** (0.0067)		0.0220*** (0.0067)	0.0323*** (0.0094)		0.0317*** (0.0094)	0.0633*** (0.0150)		0.0624*** (0.0148)
Supercenter New.		0.1117** (0.0537)	0.1088** (0.0537)		0.0438*** (0.0123)	0.0429*** (0.0125)		0.0629*** (0.0203)	0.0616*** (0.0204)		0.0903** (0.0452)	0.0877* (0.0453)
Constant	9.7845*** (0.6973)	9.9365*** (0.7247)	9.8582*** (0.7041)	−0.6277** (0.2670)	−0.5736** (0.2704)	−0.5987** (0.2664)	−0.2716 (0.3740)	−0.1937 (0.3796)	−0.2299 (0.3735)	3.0056*** (0.6530)	3.1362*** (0.6762)	3.0649*** (0.6578)
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,872	2,872	2,872	2,872	2,872	2,872	2,872	2,872	2,872	2,872	2,872	2,872
R ²	0.980	0.980	0.980	0.831	0.831	0.832	0.835	0.835	0.836	0.987	0.987	0.987

Notes. Robust Standard Errors are in parentheses. Columns (1)–(3) show results for raw MSW. Columns (4)–(12) show results for MSW normalized by population, households, and area respectively. Treated and control counties are matched using PSM on population density, employment, and wealth.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Specifically, we focus on columns (3), (6), (9), and (12) to interpret the estimated coefficients between *Supercenter_new* and *Supercenter_conv*. Our findings indicate that new launches result in a 10.88% rise in total MSW, 4.29% in *MSW per Capita*, 6.16% in *MSW per Household*, and 8.77% in *MSW per Area*. In contrast, conversions lead to increases of 6.86% in *MSW*, 2.20% in *MSW per Capita*, 3.17% in *MSW per Household*, and 6.24% in *MSW per Area*. A Wald test comparing coefficients between *Supercenter_new* and *Supercenter_conv* reveals statistically significant differences for *MSW per Capita* (F -statistic = 2.59, p -value = 0.05), and *MSW per Household* (F -statistic = 2.20, p -value = 0.07). However, the differences are not statistically significant for *MSW* (F -statistic = 0.79, p -value = 0.19) or *MSW per Area* (F -statistic = 0.39, p -value = 0.26). The results provide partial support for Hypothesis 2.

5.3. Mitigating the Supercenter Effect

In this section, we first evaluate the role of *convenience stores* and a variety of *CE channels* in ameliorating the impact of Supercenter entry on waste, as proposed in Hypothesis 3. The estimation results obtained from using Equations (4) and (5) are reported in Table 5. The estimated coefficients for the interaction terms are

consistently negative and statistically significant across all MSW metrics. Figure 5 displays changes in MSW levels across different levels of Supercenter presence and varying numbers of *convenience stores*. Furthermore, an examination of marginal effects indicates that the Supercenter effect is substantially mitigated in the greater presence of *convenience stores*. Specifically, after the entry of a Supercenter, counties with a high presence of *convenience stores*—represented by the 75th percentile—experience only a 2.75% increase in MSW. This contrasts with a 6.1% increase in counties with a median presence of *convenience stores* at the 50th percentile and a 9.3% increase in regions with a lower presence at the 25th percentile. Thus, we find the support for Hypothesis 3.

Our analysis then focused on determining whether the Supercenter effect is mitigated, at least to some degree, by the presence of these three CE channels—*CE channel (Merchandise)*, *CE channel (Perishable)*, and *CE channel (Online)*. Tables 6–8 report the estimated coefficients for the moderating effects of the three CE channels, respectively. The estimated coefficients for the interactions between the *Supercenter* entry—the number of *Supercenters* (#) and the *CE channel (Merchandise)*—are

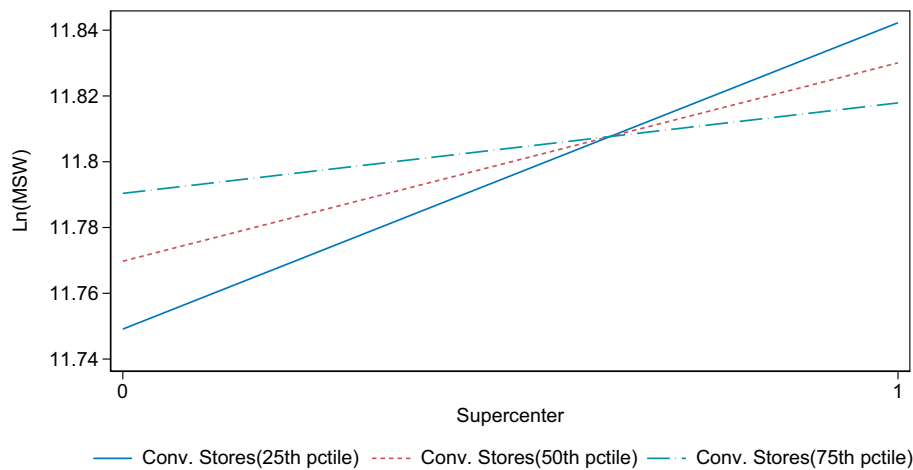
Table 5. Interaction Effect of Convenience Stores

Variable	(1) MSW	(2) MSW	(3) MSW (per capita)	(4) MSW (per capita)	(5) MSW (per Household)	(6) MSW (per Household)	(7) MSW (per Area)	(8) MSW (per Area)
<i>Median Age</i>	−0.0055 (0.0087)	−0.0055 (0.0085)	0.0069** (0.0032)	0.0070** (0.0032)	0.0047 (0.0044)	0.0047 (0.0044)	−0.0028 (0.0079)	−0.0028 (0.0077)
<i>Education</i>	−0.0007 (0.0009)	−0.0008 (0.0009)	−0.0004 (0.0005)	−0.0004 (0.0005)	−0.0008 (0.0006)	−0.0007 (0.0006)	−0.0009 (0.0009)	−0.0009 (0.0009)
<i>Online Connectivity</i>	−0.0392** (0.0170)	−0.0403** (0.0166)	−0.0199*** (0.0069)	−0.0210*** (0.0069)	−0.0299** (0.0098)	−0.0312*** (0.0098)	−0.0457*** (0.0149)	−0.0465*** (0.0147)
<i>Housing Price Index</i>	0.0782** (0.0349)	0.0830** (0.0348)	0.0484*** (0.0141)	0.0496*** (0.0141)	0.0716*** (0.0201)	0.0733*** (0.0201)	0.0993*** (0.0294)	0.1040*** (0.0294)
<i>Wealth Index</i>	0.4126*** (0.1149)	0.4129*** (0.1138)	0.1932*** (0.0482)	0.1932*** (0.0481)	0.2441*** (0.0678)	0.2442*** (0.0677)	0.4015*** (0.1100)	0.4019*** (0.1089)
<i>Supercenter</i>	0.1259*** (0.0339)		0.0628*** (0.0150)		0.0862*** (0.0208)		0.1114*** (0.0296)	
<i>Convenience Stores</i>	0.0138 (0.0315)	0.0131 (0.0311)	−0.0186 (0.0131)	−0.0199 (0.0130)	−0.0223 (0.0179)	−0.0240 (0.0178)	0.0072 (0.0275)	0.0070 (0.0273)
<i>Supercenter</i> × <i>Convenience Stores</i>	−0.0219*** (0.0076)		−0.0119*** (0.0036)		−0.0168*** (0.0050)		−0.0187*** (0.0070)	
<i>Supercenters</i> (#)		0.1292*** (0.0277)		0.0523*** (0.0124)		0.0724*** (0.0172)		0.1187*** (0.0246)
<i>Supercenters</i> (#) × <i>Convenience Stores</i>		−0.0223*** (0.0056)		−0.0098*** (0.0027)		−0.0138*** (0.0037)		−0.0202*** (0.0051)
Constant	9.7471*** (0.7184)	9.7213*** (0.7055)	−0.5515** (0.2776)	−0.5533** (0.2764)	−0.1792 (0.3877)	−0.1827 (0.3860)	2.9917*** (0.6838)	2.9650*** (0.6711)
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,869	2,869	2,869	2,869	2,869	2,869	2,869	2,869
R ²	0.980	0.980	0.832	0.832	0.835	0.836	0.987	0.987

Notes. Robust Standard Errors are in parentheses. Columns (1) and (2) show results for raw MSW. Columns (3)–(8) show results for MSW normalized by population, households, and area, respectively. Treated and control counties are matched using PSM on population density, employment, and wealth.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Figure 5. (Color online) Interaction Plot for Convenience Stores



consistently negative and statistically significant across various MSW measures. Turning to the moderating effect of *CE channel (Perishable)*, we find the estimated coefficients for the interactions between the *Supercenter* entry—both the binary indicator and the number of *Supercenters (#)*—and the *CE channel (Perishable)* are negative and statistically significant for the raw MSW measure. However, these coefficients do not maintain consistent statistical significance across normalized MSW measures, as shown in Table 7, columns (3)–(8), although they remain negative in all instances. The limited impact of food banks is likely due to the composition of their donations, which primarily come from organizations such as retailers, restaurants, distributors,

Table 6. Interaction Effect of Offline CE Channels (Merchandise)

Variable	(1) MSW	(2) MSW	(3) MSW (per capita)	(4) MSW (per capita)	(5) MSW (per Household)	(6) MSW (per Household)	(7) MSW (per Area)	(8) MSW (per Area)
Median Age	−0.0061 (0.0086)	−0.0066 (0.0083)	0.0075** (0.0032)	0.0074** (0.0032)	0.0054 (0.0044)	0.0053 (0.0044)	−0.0033 (0.0077)	−0.0038 (0.0074)
Education	−0.0007 (0.0009)	−0.0007 (0.0009)	−0.0005 (0.0005)	−0.0005 (0.0005)	−0.0008 (0.0006)	−0.0008 (0.0006)	−0.0009 (0.0009)	−0.0009 (0.0009)
Online Connectivity	−0.0394** (0.0170)	−0.0395** (0.0166)	−0.0222*** (0.0068)	−0.0229*** (0.0068)	−0.0327** (0.0097)	−0.0336*** (0.0096)	−0.0461*** (0.0146)	−0.0460*** (0.0144)
Housing Price Index	0.0772** (0.0350)	0.0809** (0.0349)	0.0491*** (0.0141)	0.0497*** (0.0142)	0.0727*** (0.0202)	0.0735*** (0.0202)	0.0985*** (0.0294)	0.1022*** (0.0294)
Wealth Index	0.4188*** (0.1142)	0.4246*** (0.1130)	0.1920*** (0.0479)	0.1936*** (0.0479)	0.2423*** (0.0673)	0.2445*** (0.0673)	0.4068*** (0.1093)	0.4124*** (0.1081)
Supercenter	0.0878*** (0.0223)		0.0419*** (0.0097)		0.0567*** (0.0137)		0.0800*** (0.0198)	
CE channel (Merchandise)	0.0132 (0.0126)	0.0196 (0.0128)	0.0006 (0.0066)	0.0020 (0.0067)	−0.0007 (0.0090)	0.0012 (0.0092)	0.0122 (0.0122)	0.0184 (0.0124)
Supercenter × CE channel (Merchandise)	−0.0194*** (0.0072)		−0.0106*** (0.0036)		−0.0150*** (0.0049)		−0.0172** (0.0069)	
Supercenters (#)		0.0932*** (0.0183)		0.0352*** (0.0081)		0.0482*** (0.0113)		0.0872*** (0.0165)
Supercenters (#) × CE channel (Merchandise)		−0.0216*** (0.0050)		−0.0091*** (0.0025)		−0.0127*** (0.0034)		−0.0199*** (0.0048)
Constant	9.7731*** (0.7004)	9.7283*** (0.6855)	−0.6363** (0.2669)	−0.6453** (0.2660)	−0.2785 (0.3749)	−0.2915 (0.3736)	2.9943*** (0.6563)	2.9505*** (0.6418)
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,872	2,872	2,872	2,872	2,872	2,872	2,872	2,872
R ²	0.980	0.980	0.831	0.831	0.836	0.836	0.987	0.987

Notes. Robust Standard Errors are in parentheses. Columns (1) and (2) show results for raw MSW. Columns (3)–(8) show results for MSW normalized by population, households, and area, respectively. Treated and control counties are matched using PSM on population density, employment, and wealth.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 7. Interaction Effect of Offline CE Channels (Perishable)

Variable	(1) MSW	(2) MSW	(3) MSW (per capita)	(4) MSW (per capita)	(5) MSW (per Household)	(6) MSW (per Household)	(7) MSW (per Area)	(8) MSW (per Area)
<i>Median Age</i>	−0.0049 (0.0085)	−0.0046 (0.0083)	0.0082*** (0.0031)	0.0083*** (0.0031)	0.0063 (0.0043)	0.0064 (0.0043)	−0.0022 (0.0076)	−0.0018 (0.0074)
<i>Education</i>	−0.0006 (0.0009)	−0.0006 (0.0009)	−0.0004 (0.0005)	−0.0004 (0.0005)	−0.0008 (0.0006)	−0.0007 (0.0006)	−0.0008 (0.0009)	−0.0008 (0.0009)
<i>Online Connectivity</i>	−0.0448*** (0.0164)	−0.0478*** (0.0161)	−0.0261*** (0.0067)	−0.0272*** (0.0067)	−0.0379*** (0.0095)	−0.0394*** (0.0095)	−0.0509*** (0.0143)	−0.0536*** (0.0141)
<i>Housing Price Index</i>	0.0738** (0.0356)	0.0755** (0.0355)	0.0469*** (0.0142)	0.0468*** (0.0142)	0.0695*** (0.0202)	0.0694*** (0.0202)	0.0960*** (0.0295)	0.0976*** (0.0295)
<i>Wealth Index</i>	0.4220*** (0.1177)	0.4220*** (0.1165)	0.1889*** (0.0486)	0.1899*** (0.0487)	0.2390*** (0.0684)	0.2398*** (0.0685)	0.4064*** (0.1122)	0.4069*** (0.1111)
<i>Supercenter</i>	0.0681*** (0.0207)		0.0233*** (0.0078)		0.0319*** (0.0109)		0.0598*** (0.0166)	
<i>CE channel (Perishable)</i>	0.0310 (0.0263)	0.0344 (0.0260)	0.0036 (0.0075)	0.0049 (0.0075)	0.0046 (0.0111)	0.0061 (0.0110)	0.0188 (0.0197)	0.0223 (0.0195)
<i>Supercenter × CE channel (Perishable)</i>	−0.0216* (0.0131)		−0.0030 (0.0050)		−0.0060 (0.0071)		−0.0162 (0.0111)	
<i>Supercenters (#)</i>		0.0621*** (0.0155)		0.0176*** (0.0060)		0.0240*** (0.0083)		0.0564*** (0.0127)
<i>Supercenters (#) × CE channel (Perishable)</i>		−0.0209** (0.0084)		−0.0046 (0.0031)		−0.0070 (0.0044)		−0.0173** (0.0070)
<i>Constant</i>	9.7397*** (0.7057)	9.7162*** (0.6948)	−0.6325** (0.2706)	−0.6360** (0.2707)	−0.2765 (0.3797)	−0.2805 (0.3798)	2.9823*** (0.6672)	2.9582*** (0.6566)
<i>County fixed effects</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Year fixed effects</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Observations</i>	2,872	2,872	2,872	2,872	2,872	2,872	2,872	2,872
<i>R²</i>	0.980	0.980	0.831	0.831	0.835	0.835	0.987	0.987

Notes. Robust Standard Errors are in parentheses. Columns (1) and (2) show results for raw MSW. Columns (3)–(8) show results for MSW normalized by population, households, and area, respectively. Treated and control counties are matched using PSM on population density, employment, and wealth.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

and growers, whereas contributions from individual consumers typically represent a smaller share. Moreover, logistical challenges associated with donating perishable goods, including the need for immediate action due to their short shelf life, make regular contributions from individuals more difficult.

Finally, the estimated coefficients for the interactions between the number of *Supercenters* (#) and the *CE channel (Online)* are consistently negative and statistically significant across various MSW measures. Although the estimated coefficients for the interactions between the *Supercenter* entry and the *CE channel (Online)* are consistently negative, they are not statistically significant. This may suggest that the mitigating effects of Craigslist on waste only become evident when the presence of more than one Supercenter contributes to a large-scale waste problem. In other words, the volume of material cycling through the system may have to be large enough for channels like Craigslist to flourish and have an impact on waste reduction. This threshold effect aligns with the prior literature on Craigslist-type of platforms, particularly from the perspective of social interactions (Zhang et al. 2015) and platform network effects (Seamans and Zhu 2014). In our context, when a substantial

surplus is generated as a result of multiple Supercenters and listed on Craigslist, it attracts more buyers and encourages additional listings, thereby enhancing the platform's ability to mitigate waste. These findings point to evidence that enhancing the presence of CE channels in various formats and forms helps alleviate the adverse environmental impact of Supercenters.

Figure 6, (a)–(c) display the variation in MSW levels with the presence of Supercenters and different levels of the three CE channels. An examination of marginal effects indicates that the Supercenter effect is substantially mitigated in the greater presence of all CE channels. Specifically, in counties with a high presence of thrift stores at the 75th percentile, there is a modest 2.5% increase in MSW. This increase is more pronounced in counties with a median presence of thrift stores at the 50th percentile, where MSW rises by 4.4%, and it escalates further to a 6.5% increase in areas with fewer thrift stores at the 25th percentile. Similarly, the high presence of food banks alleviates the increase in MSW. Counties with few food banks at the 25th percentile see a 6.8% increase in MSW, which decreases to 5.1% at the 50th percentile and drops to 3.5% at the 75th percentile. Finally, in the case of the *CE channel (Online)*, the presence of Craigslist has a

Table 8. Interaction Effect of Online CE Channels

Variable	(1) MSW	(2) MSW	(3) MSW (per capita)	(4) MSW (per capita)	(5) MSW (per Household)	(6) MSW (per Household)	(7) MSW (per Area)	(8) MSW (per Area)
Median Age	−0.0047 (0.0086)	−0.0049 (0.0085)	0.0075** (0.0032)	0.0073** (0.0032)	0.0055 (0.0044)	0.0053 (0.0044)	−0.0022 (0.0078)	−0.0024 (0.0077)
Education	−0.0007 (0.0009)	−0.0007 (0.0009)	−0.0005 (0.0005)	−0.0005 (0.0005)	−0.0008 (0.0006)	−0.0008 (0.0006)	−0.0009 (0.0009)	−0.0009 (0.0009)
Online Connectivity	−0.0467*** (0.0164)	−0.0509*** (0.0161)	−0.0252*** (0.0068)	−0.0268*** (0.0068)	−0.0373** (0.0095)	−0.0394*** (0.0096)	−0.0519*** (0.0144)	−0.0558*** (0.0143)
Housing Price Index	0.0731** (0.0354)	0.0731** (0.0353)	0.0471*** (0.0142)	0.0465*** (0.0142)	0.0695*** (0.0202)	0.0687*** (0.0202)	0.0960*** (0.0295)	0.0959*** (0.0294)
Wealth Index	0.4050*** (0.1146)	0.4005*** (0.1137)	0.1885*** (0.0481)	0.1871*** (0.0481)	0.2372*** (0.0676)	0.2353*** (0.0676)	0.3959*** (0.1098)	0.3917*** (0.1090)
Supercenter	0.0524*** (0.0152)		0.0225*** (0.0067)		0.0292*** (0.0092)		0.0475*** (0.0134)	
CE channel (Online)	0.0133 (0.0267)	0.0217 (0.0262)	−0.0031 (0.0094)	0.0009 (0.0093)	−0.0021 (0.0137)	0.0030 (0.0136)	0.0022 (0.0241)	0.0112 (0.0237)
Supercenter × CE channel (Online)	−0.0244 (0.0271)		−0.0137 (0.0109)		−0.0182 (0.0153)		−0.0141 (0.0248)	
Supercenters (#)		0.0460*** (0.0111)		0.0162*** (0.0051)		0.0210*** (0.0069)		0.0431*** (0.0101)
Supercenters (#) × CE channel (Online)		−0.0294* (0.0152)		−0.0148** (0.0063)		−0.0192** (0.0087)		−0.0238* (0.0140)
Constant	9.8451*** (0.7097)	9.8792*** (0.7001)	−0.6004** (0.2690)	−0.5804** (0.2680)	−0.2329 (0.3775)	−0.2074 (0.3764)	3.0517*** (0.6639)	3.0854*** (0.6539)
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,872	2,872	2,872	2,872	2,872	2,872	2,872	2,872
R ²	0.980	0.980	0.831	0.831	0.835	0.835	0.987	0.987

Notes. Robust Standard Errors are in parentheses. Columns (1) and (2) show results for raw MSW. Columns (3)–(8) show results for MSW normalized by population, households, and area, respectively. Treated and control counties are matched using PSM on population density, employment, and wealth.
* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

significant moderating effect. Counties without Craigslist see a 5.2% increase in MSW after Supercenter entry, whereas those with Craigslist experience only a 2.2% increase. This demonstrates that the online CE channel can effectively reduce waste by possibly providing an alternative platform for goods redistribution, thereby lessening the impact of physical Supercenter expansions on local waste levels.

6. Discussion and Conclusions

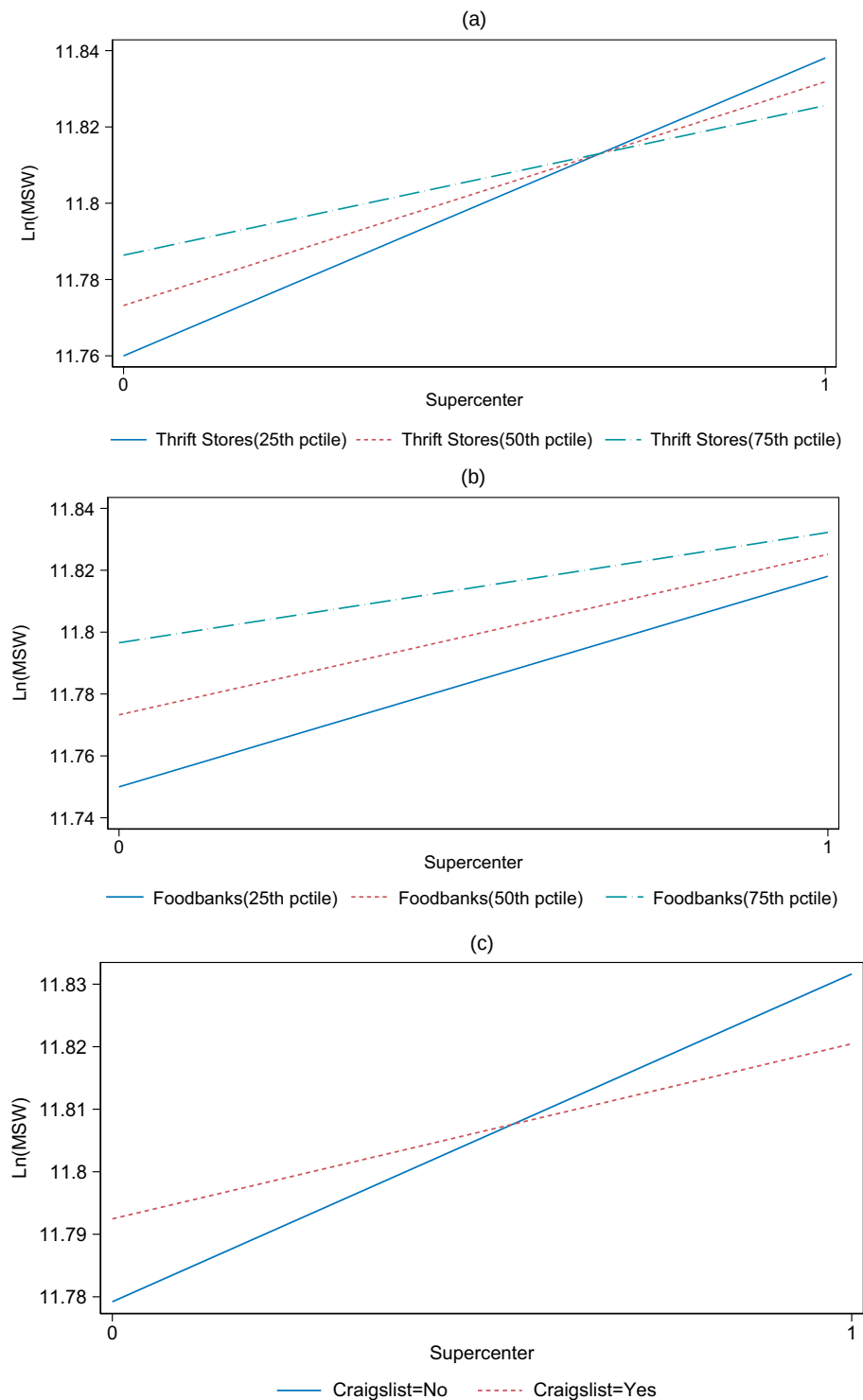
Retailers are increasingly pursuing a strategy of operating larger stores to meet the growing demand fueled by consumerism, a trend pioneered by Walmart’s expansion of its Supercenters. Although much existing literature has focused on the economic implications of such expansions, primarily analyzing their effects predominantly from an economic viewpoint, there is a significant gap in research regarding the environmental impacts of these large retail formats (Volpe and Boland 2022). Our research addresses this gap by specifically investigating the environmental effects of Supercenter expansions.

We find that the introduction of Supercenters leads to a roughly 6.97% increase in consumer waste in the

affected counties, after accounting for various demographic characteristics, time, and county fixed effects. Moreover, the magnitude of Supercenter entry depends on the type of entry. New Supercenter launches lead to a 4.29% increase in MSW per capita and a 6.16% increase in MSW per household. In contrast, conversions lead to a 2.20% increase in MSW per capita and a 3.17% increase in MSW per household. A Wald test shows that the effects of new Supercenter launches are statistically significantly higher than the effects of conversions for MSW per capita and MSW per household. These marked differences in the magnitude of impact underscore the significant influence that new Supercenters have on consumer behavior compared with store expansions.

In addition to quantifying the detrimental environmental effects of Supercenter entry, identifying strategies to alleviate these effects offers meaningful policy implications. Our research demonstrates that the environmental impact of Supercenter expansions can be substantially mitigated by a high presence of convenience stores. This underscores the significant role that convenience stores play in reducing the environmental impact of Supercenter expansions. This finding

Figure 6. (Color online) Interaction Plots for CE Channels



Notes. (a) CE Channel (Merchandise)-Thrift Stores. (b) CE Channel (Perishable)-Food Banks. (c) CE Channel (Online)-Craigslist.

is in line with the analytical insights from Belavina (2021), which noted that enhanced store accessibility leads to reduced consumer waste.

However, the trend in the retail sector has been moving toward fewer convenience stores and larger

Supercenters, driven by the pursuit of economies of scale, as observed by Jia (2008). This shift toward larger retail formats, like the expanded store formats introduced by Target Corporation (2022), requires customers to travel greater distances, thereby increasing

environmental footprints. Belavina (2021) and Cachon (2014) both emphasize that the environmental costs associated with larger store formats can outweigh the benefits of economies of scale. Furthermore, whereas Supercenters and warehouse clubs have long relied on promotions that encourage customers to buy more, often offering discounts for purchasing multiple units or large-size packages, an alternative approach is seen in dollar stores like Dollar Tree. These stores emphasize small package sizes, allowing customers to buy just what they need without generating excess waste. Indeed, they are using this as a significant differentiation point in comparison with other retail channels. Dollar Tree, for example, encourages customers to “buy just how much you need. Every time!” and offers the option to buy only as much as they want, even for online orders in some selected categories. By prioritizing small package sizes, dollar stores are providing a more sustainable and customer-friendly shopping experience. Notably, dollar stores have experienced significant growth in recent times, with Dollar General standing out for opening over 1,000 new stores in 2022, even during the COVID-19 period when other offline retailers were struggling. In the future, researchers can formally examine how package sizes at retail formats impact environmental externalities.

Given these dynamics, our findings emphasize the need for a balanced retail strategy that includes both convenience stores and Supercenters. To foster such a balance, zoning regulations can be implemented as a strategic policy tool to guide the placement and types of retail outlets, particularly by ensuring that convenience stores are accessible within residential areas. Furthermore, local governments can play a pivotal role in supporting convenience stores through various incentives such as tax breaks, reduced permit fees, and assistance in securing locations. These measures not only promote sustainability, but also ensure that consumers have convenient access to retail services, creating a more balanced and eco-friendly retail landscape.

Furthermore, our results show that the negative impacts of Supercenter entry can be substantially alleviated through the implementation of circular economy channels. For policymakers, developing and promoting such CE channels may be a low-cost solution for managing the rapid rise in consumption and consumer waste (Dhanorkar 2019). Our research lays the foundation for more research around the various other circular economy channels. For example, recycling is another important mode of disposition relevant to the circular economy (Ellen MacArthur Foundation 2012). Although abandoning or littering goods (i.e., discarding goods in a socially unacceptable way) is linked to municipal waste, we expect that more participation in CE channels will decrease the amount of waste

we throw away. For example, the state of Texas was having a tremendous problem with litter on its highways and streets. After the state launched a “Don’t Mess with Texas” campaign featuring National Football League players and popular artists to urge residents not to litter, litter was reduced by 60% (Hoyer and Macinnis 2004, p. 295). In this regard, we need more research to develop effective marketing campaigns to promote other CE channels (e.g., reuse, recycling) while discouraging littering.

Going forward, technology will also play a key role in the creation and management of CE channels. Although Craigslist was one of the first large-scale online platforms to facilitate local reuse markets, various similar markets function worldwide today. Notably, Facebook Marketplace surged to 1.2 billion active users in 2023, surpassing Craigslist due to its large existing user base.⁹ Additionally, customized product-oriented marketplaces (e.g., Poshmark, ThredUp) have also thrived. Second-hand product transactions based on online platforms have recently become a trend, especially among the younger generations, in many counties (Lee 2022). Although big-box retail might be here to stay, this may be an opportunity for managers to find ways for promoting the use of CE channels (and other related proenvironmental efforts, e.g., reuse, recycling) through media campaigns. Notably, major retailers like Walmart and Target have started engaging with platforms like ThredUp to promote reuse and secondhand shopping (Hale 2022, Target Corporation 2022).

Our study has some limitations. First, county-level consumer waste data are not widely available and may not be collected by all states, unless mandated by the state environmental agency. Most states in the United States either do not publish annual data or publish only state-level estimates, which preclude a temporal analysis using the above-described methodology. Currently, there is no central repository such as the U.S. Census Bureau that collects such data. Thus, we limit our analysis to four states (i.e., California, Florida, Minnesota, and North Carolina) that publish annual county-level consumer waste data. Second, Craigslist primarily expanded into larger metropolitan areas in the early stage, many of which may already have a Walmart Supercenter. This expansion pattern introduces potential challenges in identifying a suitable control group of counties that lack either a Walmart Supercenter or Craigslist presence. Furthermore, because of lack of data on Craigslist’s historical posts related to personal item exchanges and sales, we are not able to investigate the mechanisms underlying the mitigating effects of Craigslist on consumer waste. Third, our study relies on a natural experiment using observational data, which may not achieve the level of definitive causality that randomized controlled experiments can provide.

Consequently, the mitigating effects of convenience stores and CE channels on the impact of Supercenters may not represent conclusive causal relationships. Future research could explore additional strategies to mitigate the increase in consumer waste due to Supercenters. Notwithstanding these shortcomings, this study represents one of the first attempts to quantify the effects of Supercenters on consumer waste, as well as factors that can mitigate this effect.

Acknowledgments

The authors thank the department editor, Anita L. Carson, for the valuable guidance. The authors also thank the anonymous associate editor and reviewers, who provided extremely constructive feedback.

Endnotes

- ¹ Store counts and growth rates are calculated based on store information disclosed in Walmart annual reports (1999–2022) <https://stock.walmart.com/financials/annual-reports/> (accessed February 11, 2023).
- ² We report the year of the Supercenter entry for each county in our data sample in Table A2 of Online Appendix A.
- ³ See <https://www.census.gov/data/tables/2017/econ/e-stats/2017-e-stats.html> (accessed March 27, 2023).
- ⁴ See <http://www.calrecycle.ca.gov/LGCentral/DataTools/Reports/> (accessed November 18, 2019).
- ⁵ See <https://floridadep.gov/waste/waste-reduction/content/publications-links-and-further-recycling-assistance-and-information> (accessed November 18, 2019).
- ⁶ See <https://www.pca.state.mn.us/waste/recycling-minnesota-score-report> (accessed November 18, 2019).
- ⁷ See <http://deq.nc.gov/about/divisions/waste-management/waste-management-rules-data> (accessed November 18, 2019).
- ⁸ Source: North American Industry Classification System: 445120 Convenience Stores (2017) U.S. Census Bureau. Retrieved from <https://www.census.gov/naics/?input=445120&year=2017&details=445120>.
- ⁹ See <https://fortune.com/2024/06/01/mark-zuckerberg-facebook-marketplace-e-commerce-amazon-gen-z/> (accessed June 4, 2024).

References

- Agrawal VV, Atasu A, Van Wassenhove LN (2019) OM forum—New opportunities for operations management research in sustainability. *Manufacturing Service Oper. Management* 21(1):1–12.
- Ailawadi KL, Zhang J, Krishna A, Kruger MW (2010) When Wal-Mart enters: How incumbent retailers react and how this affects their sales outcomes. *J. Marketing Res.* 47(4):577–593.
- Akkaş A, Gaur V (2022) OM forum—Reducing food waste: An operations management research agenda. *Manufacturing Service Oper. Management* 24(3):1261–1275.
- Akkaş A, Honhon D (2022) Shipment policies for products with fixed shelf lives: Impact on profits and waste. *Manufacturing Service Oper. Management* 24(3):1611–1629.
- Alev I, Agrawal VV, Atasu A (2020) Extended producer responsibility for durable products. *Manufacturing Service Oper. Management* 22(2):364–382.
- Arcidiacono P, Ellickson PB, Mela CF, Singleton JD (2020) The competitive effects of entry: Evidence from Supercenter expansion. *Amer. Econom. J. Appl. Econom.* 12(3):175–206.
- Aschemann-Witzel J, De Hooge I, Normann A (2016) Consumer-related food waste: Role of food marketing and retailers and potential for action. *J. Internat. Food Agribusiness Marketing* 28(3):271–285.
- Aschemann-Witzel J, Jensen JH, Jensen MH, Kulikovskaja V (2017) Consumer behaviour towards price-reduced suboptimal foods in the supermarket and the relation to food waste in households. *Appetite* 116:246–258.
- Basker E (2005) Selling a cheaper mousetrap: Wal-Mart's effect on retail prices. *J. Urban Econom.* 58(2):203–229.
- Basker E, Noel M (2009) The evolving food chain: Competitive effects of Wal-Mart's entry into the supermarket industry. *Econom. Management Strategy* 18(4):977–1009.
- Belavina E (2021) Grocery store density and food waste. *Manufacturing Service Oper. Management* 23(1):1–18.
- Belavina E, Girotra K, Kabra A (2017) Online grocery retail: Revenue models and environmental impact. *Management Sci.* 63(6):1781–1799.
- Belk RW, Sherry JF Jr, Wallendorf M (1988) A naturalistic inquiry into buyer and seller behavior at a swap meet. *J. Consumer Res.* 14:449.
- Bertrand M, Duflo E, Mullainathan S (2004) How much should we trust differences-in-differences estimates? *Quart. J. Econom.* 119(1):249–275.
- Bonanno A, Goetz SJ (2012) WalMart and local economic development: A survey. *Econom. Development Quart.* 26(4):285–297.
- Brynjolfsson E, Hu Y, Simester D (2011) Goodbye pareto principle, hello long tail: The effect of search costs on the concentration of product sales. *Management Sci.* 57(8):1373–1386.
- Cachon GP (2014) Retail store density and the cost of greenhouse gas emissions. *Management Sci.* 60(8):1907–1925.
- Caro F, Kök AG, Martínez-de-Albéniz V (2020) The future of retail operations. *Manufacturing Service Oper. Management* 22(1):47–58.
- Chan J, Ghose A, Seamans R (2016) Internet and racial hate crime: Offline spillovers from online access. *MIS Quart.* 40(2):381–403.
- Chen J, Esteban S, Shum M (2013) When do secondary markets harm firms? *Amer. Econom. Rev.* 103(7):2911–2934.
- Chenarides L, Çakır M, Richards TJ (2024) Dynamic model of entry: Dollar stores. *Amer. J. Agricultural Econom.* 106(2):852–882.
- Chernev A (2003) Product assortment and individual decision processes. *J. Personality Soc. Psych.* 85(1):151–162.
- Courtemanche C, Carden A (2011) Supersizing supercenters? The impact of Walmart Supercenters on body mass index and obesity. *J. Urban Econom.* 69(2):165–181.
- Courtemanche C, Carden A, Zhou X, Ndirangu M (2019) Do Walmart Supercenters improve food security? *Appl. Econom. Perspect. Policy* 41(2):177–198.
- Dhanorkar S (2019) Environmental benefits of internet-enabled C2C closed-loop supply chains: A quasi-experimental study of Craigslist. *Management Sci.* 65(2):660–680.
- Dhanorkar S, Muthulingam S (2020) Do e-waste laws create behavioral spillovers? Quasi-experimental evidence from California. *Product. Oper. Management* 29(7):1738–1766.
- Ellen MacArthur Foundation (2012) Towards the Circular Economy Vol. 1: An economic and business rationale for an accelerated transition. Accessed November 18, 2019, <https://ellenmacarthurfoundation.org/towards-the-circular-economy-vol-1-an-economic-and-business-rationale-for-an>.
- Ellison G, Glaeser EL, Kerr WR (2010) What causes industry agglomeration? Evidence from coagglomeration patterns. *Amer. Econom. Rev.* 100(3):1195–1213.
- EPA (2011) Municipal solid waste generation, recycling, and disposal in the United States: Facts and figures for 2011. Accessed June 4, 2025, https://archive.epa.gov/epawaste/nonhaz/municipal/web/pdf/mswcharacterization_508_053113_fs.pdf.

- EPA (2013) Waste management hierarchy. Accessed June 4, 2025, <https://www.epa.gov/smm/sustainable-materials-management-non-hazardous-materials-and-waste-management-hierarchy>.
- EPA (2014) Advancing sustainable materials management: 2014 fact sheet. Accessed November 18, 2019, https://www.epa.gov/sites/production/files/2016-11/documents/2014_smmfactsheet_508.pdf.
- EPA (2016) Municipal solid waste. Accessed November 18, 2019, <https://archive.epa.gov/epawaste/nonhaz/municipal/web/html/>.
- EPA (2020) Advancing sustainable materials management: 2018 fact sheet: Assessing trends in materials generation and management in the United States. Accessed June 4, 2024, https://www.epa.gov/sites/default/files/2021-01/documents/2018_ff_fact_sheet_dec_2020_fnl_508.pdf.
- Farley JU, Ring LW (1966) A stochastic model of supermarket traffic flow. *Oper. Res.* 14(4):555–567.
- Flamand T, Ghoniem A, Maddah B (2023) Store-wide shelf-space allocation with ripple effects driving traffic. *Oper. Res.*
- Fremstad A (2017) Does Craigslist reduce waste? Evidence from California and Florida. *Ecological Econom.* 132:135–143.
- Ghose A, Telang R, Krishnan R (2005) Effect of electronic secondary markets on the supply chain. *J. Management Inform. Systems* 22(2): 91–120.
- Glaeser E (2012) *Triumph of the City: How Our Greatest Invention Makes Us Richer, Smarter, Greener, Healthier, and Happier* (Penguin Books, New York).
- Gourville J, Soman D (2002) Pricing and the psychology of consumption. *Harvard Bus. Rev.* 80(9):90–96.
- Greenwood BN, Agarwal R (2016) Matching platforms and HIV incidence: An empirical investigation of race, gender, and socioeconomic status. *Management Sci.* 62(8):2281–2303.
- Hale K (2022) Target taps into \$51 billion second hand clothing market with ThredUp deal. *Forbes* (May 12), <https://www.forbes.com/sites/korihale/2022/05/12/target-taps-into-51-billion-second-hand-clothing-market-with-thredup-deal/?sh=29a9f66b2da3>.
- Halebsky S (2009) *Small Towns and Big Business: Challenging Wal-Mart Superstores* (Lexington Books).
- Hausman A (2000) A multi-method investigation of consumer motivations in impulse buying behavior. *J. Consumer Marketing* 17(5): 403–426.
- Hausman J, Leibtag E (2007) Consumer benefits from increased competition in shopping outlets: Measuring the effect of Wal-Mart. *J. Appl. Econometrics* 22(7):1157–1177.
- Heckman JJ, Ichimura H, Todd PE (1997) Matching as an econometric evaluation estimator: Evidence from evaluating a job training programme. *Rev. Econom. Stud.* 64(4):605–654.
- Holmes TJ (2011) The diffusion of Wal-Mart and economies of density. *Econometrica* 79(1):253–302.
- Homonoff TA (2018) Can small incentives have large effects? The impact of taxes versus bonuses on disposable bag use. *Amer. Econom. J. Econom. Policy* 10(4):177–210.
- Hoyer WD, Macinnis DJ (2004) *Customer Behavior*, 3rd ed. (Houghton Mifflin Company, Boston).
- Hwang M, Park S (2016) The impact of Walmart supercenter conversion on consumer shopping behavior. *Management Sci.* 62(3):817–828.
- Iyengar SS, Lepper MR (2000) When choice is demotivating: Can one desire too much of a good thing? *J. Personality Soc. Psych.* 79(6):995.
- Jia P (2008) What happens when Wal-Mart comes to town: An empirical analysis of the discount retailing industry. *Econometrica* 76(6):1263–1316.
- Korhonen J, Honkasalo A, Seppälä J (2018) Circular economy: The concept and its limitations. *Ecological Econom.* 143:37–46.
- Lee D (2022) Lacoste and YSL to Burberry on a budget: A trend for second-hand designer clothes is taking South Korea by storm. *South China Morning Post* (February 2), <https://www.scmp.com/week-asia/lifestyle-culture/article/3165481/lacoste-and-ysl-burberry-budget-trend-second-hand>.
- Leibtag ES (2006) The impact of big-box stores on retail food prices and the consumer price index. Economic Research Report No. 33, Economic Research Service, U.S. Department of Agriculture, Washington, DC.
- Long X, Nasiry J (2022) Sustainability in the fast fashion industry. *Manufacturing Service Oper. Management* 24(3):1276–1293.
- Lowrey JD, Richards TJ, Hamilton SF (2023) Food donations, retail operations, and retail pricing. *Manufacturing Service Oper. Management* 25(2):792–810.
- Martin J, Nenycz-Thiel M, Dawes J, Tanusondjaja A, Cohen J, McColl B, Trinh G (2020) Fundamental basket size patterns and their relation to retailer performance. *J. Retailing Consumer Services* 54:102032.
- Matsa DA (2011) Competition and product quality in the supermarket industry. *Quart. J. Econom.* 126(3):1539–1591.
- Miranda ML, Everett JW, Blume D, Roy BA Jr (1994) Market-based incentives and residential municipal solid waste. *J. Policy Anal. Management* 13(4):681–698.
- Netz B, Davidson OR, Bosch PR, Dave R, Meyer LA (2007) Climate change 2007: Mitigation. Contribution of Working Group III to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change. Summary for Policymakers. Intergovernmental Panel on Climate Change (IPCC), Geneva.
- Plambeck E, Wang Q (2009) Effects of e-waste regulation on new product introduction. *Management Sci.* 55(3):333–347.
- Richards TJ, Hamilton SF (2018) Food waste in the sharing economy. *Food Policy* 75:109–123.
- Rosenbaum PR, Rubin DB (1983) The central role of the propensity score in observational studies for causal effects. *Biometrika* 70(1): 41–55.
- Rosenbaum PR, Rubin DB (1985) Constructing a control group using multivariate matched sampling methods that incorporate the propensity score. *Amer. Statistician* 39(1):33–38.
- Rubin DB (2001) Using propensity scores to help design observational studies: Application to the tobacco litigation. *Health Services Outcomes Res. Methodology* 2(3):169–188.
- Schulze G (2016) *Growth Within: A Circular Economy Vision for a Competitive Europe* (Ellen MacArthur Foundation and the McKinsey Center for Business and Environment), 1–22.
- Seamans R, Zhu F (2014) Responses to entry in multi-sided markets: The impact of Craigslist on local newspapers. *Management Sci.* 60(2):476–493.
- Sherry JF (1990) A sociocultural analysis of a Midwestern American flea market. *J. Consumer Res.* 13–30.
- Singh VP, Hansen KT, Blattberg RC (2006) Market entry and consumer behavior: An investigation of a Wal-Mart supercenter. *Marketing Sci.* 25(5):457–476.
- Stahel WR (2016) The circular economy. *Nature* 531(7595):435.
- Suslow VY (1986) Estimating monopoly behavior with competitive recycling: An application to Alcoa. *RAND J. Econom.* 389–403.
- Target Corporation (2022) Target debuts new larger-format store featuring modern design and expanded space to fuel popular same-day fulfillment services. Accessed February 11, 2023, <https://corporate.target.com/press/release/2022/11/target-debuts-new-larger-format-store-featuring-mo>.
- Taylor S, Todd P (1995) Understanding household garbage reduction behavior: A test of an integrated model. *J. Public Policy Marketing* 14(2):192–204.
- Tchobanoglous G, Kreith F (2002) *Handbook of Solid Waste Management* (McGraw-Hill Education, New York).
- Tsiros M, Heilman CM (2005) The effect of expiration dates and perceived risk on purchasing behavior in grocery store perishable categories. *J. Marketing* 69(2):114–129.

- Vergara SE, Tchobanoglous G (2012) Municipal solid waste and the environment: A global perspective. *Annu. Rev. Environment Resources* 37(1):277–309.
- Volpe R, Boland MA (2022) The economic impacts of Walmart Supercenters. *Annual Rev. Resource Econom.* 14:43–62.
- Volpe RJ III, Lavoie N (2008) The effect of Wal-Mart supercenters on grocery prices in New England. *Rev. Agricultural Econom.* 30(1):4–26.
- Waldman M (2003) Durable goods theory for real world markets. *J. Econom. Perspect.* 17(1):131–154.
- Williams H, Lindström A, Trischler J, Wikström F, Rowe Z (2020) Avoiding food becoming waste in households—The role of packaging in consumers’ practices across different food categories. *J. Cleaner Production* 265:121775.
- Xu K, Chan J, Ghose A, Han SP (2017) Battle of the channels: The impact of tablets on digital commerce. *Management Sci.* 63(5):1469–1492.
- Zhang J, Liu Y, Chen Y (2015) Social learning in networks of friends versus strangers. *Marketing Sci.* 34(4):573–589.
- Zhu T, Singh V (2009) Spatial competition with endogenous location choices: An application to discount retailing. *QME* 7:1–35.